

**2024**

*Time : 2 hours*

*Full Marks : 40*

*Candidates are required to give their answers  
in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Answer all sections as directed.*

**Section-A**

1. Answer any **four** questions of the following :  $6 \times 4 = 24$
- (a) State and prove Bayes' theorem.
  - (b) Define arithmetic mean with properties.
  - (c) How will you calculate median in case of grouped data?

- (d) Discuss about Poisson distribution.
- (e) Discuss about binomial distribution.
- (f) What do you understand by correlation between two variables?

### Section-B

2. Answer any **four** questions of the following :

$$4 \times 4 = 16$$

- (a) A box contains 3 red, 4 white and 5 black balls. Two balls are drawn at random from the box. Find the probability that both the balls are red or white.
- (b) The following table gives the diastolic blood pressure of 250 men. The readings were made to the nearest millimeter and the central value of each group is given by:

|                     |    |    |    |    |     |    |    |    |
|---------------------|----|----|----|----|-----|----|----|----|
| Blood pressure (mm) | 60 | 65 | 70 | 75 | 80  | 85 | 90 | 95 |
| No. of men          | 4  | 5  | 31 | 39 | 114 | 30 | 25 | 2  |

Calculate the mean and median from the data.

- (c) Calculate the Correlation Coefficient for the following heights (in inches) of father (X) and their son (Y):

|    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|
| X: | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 |
| Y: | 67 | 68 | 65 | 72 | 72 | 69 | 65 | 71 |

- (d) Define correlation coefficient and how it differs from regression.
- (e) There are two bags I and II. Bag I contains 3 white and 4 black balls and bag II contains 5 white and 6 black balls. One ball is drawn at random from one of the bags and is found to be white. Find

the probability that it was drawn from bag I.

- (f) A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability of the following events :
- (i) The card may be diamond
  - (ii) The card may be a black king

————— x —————